

**YEAR 9 SCIENCE  
CHEMICALS SCIENCES  
TEST 2 - CHEMICAL REACTIONS**

NAME: SOLUTIONS

CLASS: \_\_\_\_\_

Mark: 33

Chemical reactions, including combustion and the reactions of acids, are important in both non-living and living systems and involve energy transfer.

**Multiple Choice** Write the answer to each question in the appropriate box at right.

1. Which of the following word equations correctly explains an acid reacting with a metal?

- (a) zinc + hydrochloric acid → zinc chloride + hydrogen
- (b) magnesium + sulphuric acid → magnesium chloride + hydrogen
- (c) magnesium + hydrochloric acid → carbon dioxide + hydrogen
- (d) zinc + hydrochloric acid → zinc chloride + carbon dioxide

2. Which of the following word equations correctly explains an acid reacting with a carbonate?

- (a) hydrochloric acid + calcium carbonate → calcium chloride + water + carbon dioxide
- (b) hydrochloric acid + calcium carbonate → calcium chloride + water + hydrogen
- (c) sulphuric acid + calcium carbonate → calcium chloride + water + carbon dioxide
- (d) sulphuric acid + calcium carbonate → calcium sulphate + hydrogen + carbon dioxide

3. Which of the following is **not** a property of acids?

- (a) React with some metals to produce hydrogen gas.
- (b) Turn litmus red.
- (c) Sweet taste.
- (d) Sour taste.

4. Which of the following is a **base**?

- (a) NaCl (sodium chloride)
- (b) NaOH (sodium hydroxide)
- (c) CH<sub>3</sub>OH (methanol)
- (d) HCl (hydrochloric acid)

5. Which of the following could be the pH for a pool that had turned **slightly acidic**?

- (a) 2
- (b) 6.5
- (c) 7.5
- (d) 7

| QUESTION | ANSWER |
|----------|--------|
| 1        | A      |
| 2        | A      |
| 3        | C      |
| 4        | B      |
| 5        | B      |
| 6        | A      |
| 7        | A      |
| 8        | A      |
| 9        | A      |
| 10       | B      |
| 11       | B      |
| 12       | C      |

6. Which of the following is the correct **general equation** to describe the reaction of an acid and a base?
- (a) acid + base  $\rightarrow$  a salt + water
  - (b) acid + base  $\rightarrow$  metal hydroxide + carbon dioxide
  - (c) acid + base  $\rightarrow$  carbon dioxide + water
  - (d) acid + base  $\rightarrow$  hydrogen + water
7. Magnesium hydroxide can be taken to settle stomach acids. The reason why it is useful for this is because:
- (a) the magnesium hydroxide neutralises the stomach acid.
  - (b) the magnesium hydroxide is not very acidic so it won't make your stomach more acidic.
  - (c) magnesium hydroxide is a base so it will turn the stomach acid into a salt.
  - (d) stomach acid will turn into a base when the magnesium hydroxide is taken.
8. Which of the following is most likely to be the pH of lemon juice?
- (a) 3
  - (b) 7
  - (c) 10
  - (d) 14
9. Which of the following will turn universal indicator **red**?
- (a) Vinegar.
  - (b) Sodium hydroxide.
  - (c) Distilled water.
  - (d) Glucose solution.
10. A chemist tells you that a substance she has tested has a pH of 1. From this you know:
- (a) its colour.
  - (b) that it is a strong acid.
  - (c) that it is a weak base.
  - (d) that it is concentrated.
11. Which of the following facts about toothpaste explain why it helps reduce tooth decay?
- (a) It has a minty taste.
  - (b) It is slightly basic.
  - (c) It contains fluoride.
  - (d) It is abrasive.
12. The test for determining whether a colourless, odourless gas is hydrogen is called the:
- (a) glowing splint test.
  - (b) pH test.
  - (c) pop test.
  - (d) limewater test.

### Short Answer

Write the answer to the questions in the spaces provided.

1. List **two** characteristics of acids and bases. (Any 2 for each - 1 mark each.)

#### Acids

- Produce  $H^+$  ions in solution.
- Have a sour taste.
- Turn blue litmus red.

#### Bases

- Produce  $OH^-$  ions in solution.
- Any metal oxide or hydroxide.
- Have a soapy feel.
- Have a bitter taste.
- Turn red litmus blue.

(4)

2. Classify the following compounds as **ACID**, **BASE** or **SALT**.

$H_2SO_4$     $Mg(OH)_2$    sodium chloride    $Al_2O_3$     $HNO_3$     $CuSO_4$    calcium oxide    $HCl$

| ACID                          | BASE                                                                 | SALT                        |
|-------------------------------|----------------------------------------------------------------------|-----------------------------|
| $H_2SO_4$<br>$HNO_3$<br>$HCl$ | $Mg(OH)_2$<br>$Al_2O_3$<br>calcium oxide<br>(½ mark off each error.) | sodium chloride<br>$CuSO_4$ |

(4)

3. Complete the following table, giving **estimates on the pH where necessary**, the **colour of the universal indicator** and whether the substance is a **strong/weak acid**, a **strong/weak base** or **neutral**.

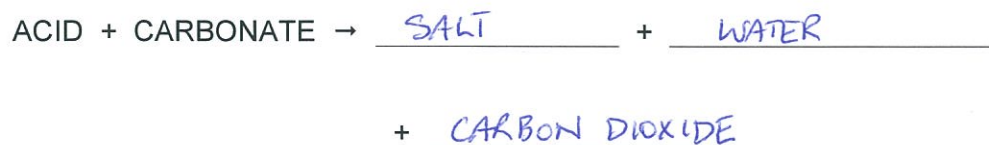
| SUBSTANCE                           | pH  | UNIVERSAL INDICATOR (colour) | ACID / BASE/ NEUTRAL (strong / weak) |
|-------------------------------------|-----|------------------------------|--------------------------------------|
| vinegar                             | 5   | orange / yellow              | weak acid                            |
| soda water                          | 3   | light red                    | weak acid                            |
| 2.00 M $H_2SO_4$ (sulphuric acid)   | 0/1 | red                          | strong acid                          |
| 1.00 M $Na_2SO_4$ (sodium sulphate) | 8   | light blue                   | weak base                            |
| sugar solution                      | 7   | green                        | neutral                              |
| yoghurt                             | 5/6 | yellow                       | weak acid                            |

↑  
accept either.

(½ mark off each error.)

(6)

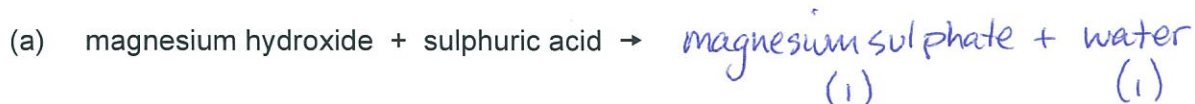
4. Complete the following general word equations for acid reactions.



( $\frac{1}{2}$  mark off each error.)

(3)

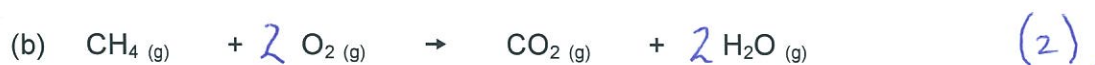
5. Complete the following **WORD EQUATIONS** for acid reactions, naming the salt produced and other products.



(4)

6. **Optional**

Balance the following equations.



(4)